**Pure Mathematics Year 2**

Chapter 6

Trigonometric Functions

## Starter — Prerequisites



1. Sketch $y = \sin x$ for $-180° \leq x \leq 180°$. Use it to solve:

a. $\sin x = 0.8$

b. $\sin x = -0.4$.

a

1 sketch

2. Prove $\frac{1}{\sin x \cos x} - \frac{1}{\tan x} = \tan x$.

2

3. Solve $3\sin^2(2x) = 1, 0 \leq x \leq 2\pi$.

3

## Chapter Overview



This chapter is very similar to the trigonometry chapters in Year 1. The only difference is that new trig functions: sec, cosec and cot, are introduced.

**1. Reciprocal functions and their graphs**

Draw $y = \operatorname{cosec} x$ for $0 \leq x < 2\pi$.

**2. Solving**

Solve $2\operatorname{cosec}^2 x + \cot x = 5$, $0 \leq x < 2\pi$, to 3 s.f.

**3. Proving**

Prove $\sec x - \cos x \equiv \sin x \tan x$.

**4. Inverse functions: domains and ranges**

Understand and sketch arcsin, arccos and arctan.

**Small-angle connection**

Show that, for small $\theta$,

$\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1$.

Here $\theta$ is in radians.

## A new member of the trig family...



$\cos x$

You already know sine, cosine and tangent.

$\cos^2 x = (\cos x)^2$

The latter notation is useful in differentiation (Chapter 9).

$\cos^{-1} x = \arccos x$

Here $-1$ denotes the inverse function, not a reciprocal power. This historical notation differs from $\cos^2 x$.

$\sec x = \dfrac{1}{\cos x}$

We have convenient names for reciprocals of trigonometric functions.

$\sec x = \dfrac{1}{\cos x}$

Short for secant: “sehk” or “sea-kant”.

$\operatorname{cosec} x = \dfrac{1}{\sin x}$

Short for cosecant.

$\cot x = \dfrac{\cos x}{\sin x}$

Also $\cot x = 1/\tan x$ wherever both sides are defined.

Cotangent (“cot” rhymes with “pot”).

**Tip**

Look at the third letter: the c in sec reminds you of 1 over cos.

In proofs, use $\cot x = \cos x/\sin x$.

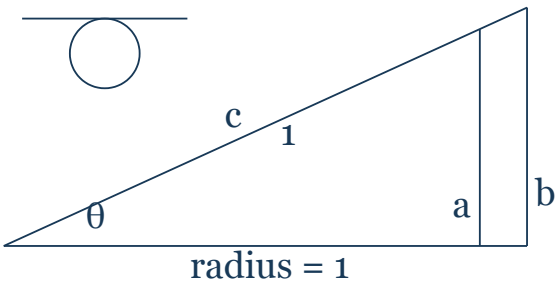


Is ‘tangent’ (tan) in trigonometry related to the ‘tangent’ of a circle?

A tangent is a trigonometric function (which inputs an angle and gives you the ratio between the opposite and adjacent sides of a right-angled triangle), but also a line which touches a circle. Are they related?

$\tan\theta$ = opposite/adjacent.

Just as ‘radius’ can refer either to the line itself or its length, we have names for other special lines, which can also refer to their lengths:



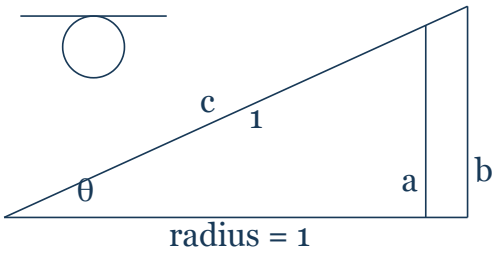

A common myth is that sin, cos and tan are the ‘core’ trigonometric functions.

Actually, they’re sin, tan and *sec*!

A tangent is a line which touches the circle. We’re interested in the length just between the touching point and where it meets the secant.

Sine (sort of) comes from the word for ‘bowstring’. It refers to half the line if we doubled up the arc and connected the two ends.

A secant (shortened to ‘sec’) is a line which cuts the circle. In a trig setting, we’re interested in the length from the circle centre to where it meets the tangent.




Suppose we let the radius of the sector be 1. Then if we define $\sin\theta = \frac{opp}{hyp}$ and $\tan\theta = \frac{opp}{adj}$ and $\sec\theta = \frac{hyp}{adj}$ then this conveniently gives us:

$\sin\theta = \frac{a}{1} = a$ $\quad\quad$ $\tan\theta = \frac{b}{1} = b$

$\sec\theta = \frac{c}{1} = c$

i.e. $\sin\theta$, $\tan\theta$ and $\sec\theta$ actually give us the lengths of the sine, tangent and secant respectively.

And hence $\tan\theta$ and the length of the 'tangent' are clearly linked!

"So if sin, tan and sec are the 'core' trigonometric functions, where does *cos* come into the fray?"

"cosine" (cos), "cotangent" (cot) and "cosecant" ( cosec) are the 'complementary' trig functions. Complementary angles add to 90°.

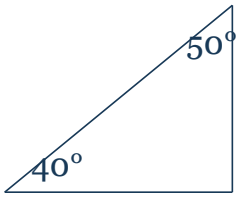
The two smaller angles in a right-angled triangle are clearly complementary.

The complementary sine (cosine) of an angle is by definition the sine of the complementary angle, e.g. $\cos(40°) = \sin(50°)$. Thus:

$\cos(\theta) = \sin(90° - \theta)$ $\quad$ $\operatorname{cosec}(\theta) = \sec(90° - \theta)$

$\cot(\theta) = \tan(90° - \theta)$

The complementary definition gives $\cot\theta = \tan(\pi/2 - \theta) = \cos\theta/\sin\theta$. Interchanging opposite and adjacent leads to $1/\tan\theta$ wherever that reciprocal is defined. The quotient $\cos\theta/\sin\theta$ also defines cotangent where $\cos\theta = 0$.






You have a calculator in A Level exams, but it's good however to know how to calculate certain values yourself if needed.

1. $\cot\frac{\pi}{4}$

1

2. $\sec\frac{\pi}{4}$

2

3. $\operatorname{cosec}\frac{\pi}{3}$

3

4. $\cot\frac{\pi}{6}$

4

5. $\operatorname{cosec}\frac{5\pi}{6}$

5

6. $\cot\frac{\pi}{3}$

6

7. $\sec\frac{\pi}{6}$

7

8. $\operatorname{cosec}\frac{\pi}{2}$

8

9. $\sec\frac{5\pi}{3}$

9

4. Prove $\operatorname{cosec}(\pi - x) = \operatorname{cosec} x$.

4

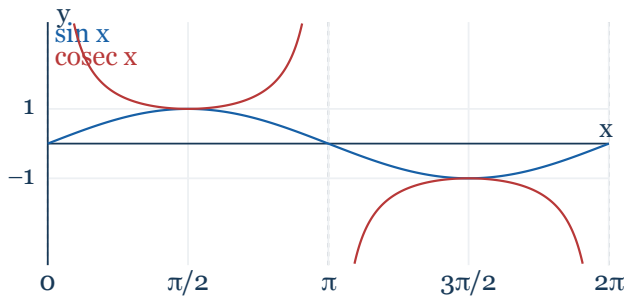
5. Show $\cot 30^\circ \sec 30^\circ = 2$.

5

6. Write $\operatorname{cosec} \frac{2\pi}{3} + \sec \frac{2\pi}{3} = a + b\sqrt{3}$.

6

## Sketches — cosec **Luisa · 10**




Draw $y = \operatorname{cosec} x$, starting with $y = \sin x$.

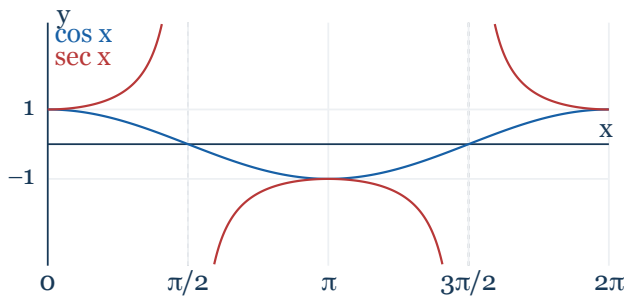
To draw a graph of $y = \operatorname{cosec} x$, start with a graph of $y = \sin x$, then consider what happens when we reciprocate each $y$ value.

It touches here because the reciprocal of 1 is 1.

cosec isn't defined for multiples of $\pi$ because we can't divide by 0. We get an asymptote.

Reciprocating preserves sign. When we divide by a small positive number, we get a very large positive number.

Domain: $x \neq n\pi,\ n \in \mathbb{Z}$.
Range: $y \leq -1$ or $y \geq 1$.

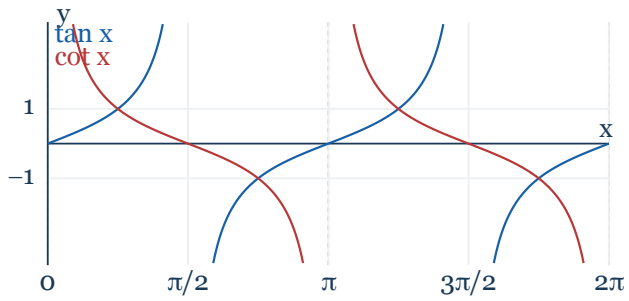



Draw $y = \sec x$, starting with $y = \cos x$.

Domain: $x \neq \frac{\pi}{2} + n\pi,\ n \in \mathbb{Z}$.

Range: $y \leq -1$ or $y \geq 1$.






Draw $y = \cot x$, starting with $y = \tan x$.

**Extra insight**

The relation $\cot\theta = \tan(\pi/2 - \theta)$ reflects the tangent graph in $\theta = \pi/4$. It is defined when $\sin\theta \neq 0$. The input angles $\theta$ and $\pi/2 - \theta$ are symmetric about $\pi/4$.

Domain: $x \neq n\pi,\ n \in \mathbb{Z}$.

Range: $y \in \mathbb{R}$.

## Example

**Luisa · 13**

[Textbook]

a. Sketch the graph of $y = 4\,\text{cosec}\,x, \quad -\pi \leq x \leq \pi$.

b. On the same axes, sketch the line $y = x$.

c. State the number of solutions to the equation $4\,\text{cosec}\,x - x = 0, \quad -\pi \leq x \leq \pi$

c

a–b

## Test Your Understanding

**Luisa · 14**

Sketch $y = -1 + \sec 2x$ in the interval $0 \leq x < 360°$.

Draw the transformations stage by stage, unless you feel comfortable doing multiple transformations at once.

1. Original

2. Compress

3. Translate

1a. Sketch, for $-540^\circ \leq \theta \leq 540^\circ$:

i. $y = \sec\theta$ ii. $y = \operatorname{cosec}\theta$ iii. $y = \cot\theta$.

i

ii

iii



2a. Sketch $y = \cot x$ and $y = -x$, $-\pi \leq x \leq \pi$, on the same axes.

2b. Deduce the number of solutions of $\cot x + x = 0$ in this interval.

a

b

6a. Describe the relationships between these pairs:

i. $\tan(\theta + \pi/2)$, $\tan\theta$.

ii. $\cot(-\theta)$, $\cot\theta$.

iii. $\operatorname{cosec}(\theta + \pi/4)$, $\operatorname{cosec}\theta$.

iv. $\sec(\theta - \pi/4)$, $\sec\theta$.

a

6b. Considering the graphs of the four transformed functions, which pairs are equal?

b sketch

b

b second pair



9a. Sketch $y = 3 + 5\operatorname{cosec} x$, $-2\pi \leq x \leq 2\pi$.

a

a sketch

9b. Deduce the values of $k$ for which $3 + 5\operatorname{cosec} x = k$ has no solutions.

b

10a. Sketch $y = 1 + 2\sec\theta$, $-\pi \leq \theta \leq 2\pi$.

a

10b. State where the gradient is zero.

b

10c. Deduce the maximum and minimum of $1/(1 + 2\sec\theta)$ and the smallest positive $\theta$ at which each occurs.

c

## Using sec, cosec and cot **Luisa · 20**

Questions in the exam usually come in two flavours: (a) 'provey' questions requiring to prove some identity and (b) 'solvey' questions.

Tip 1: Get everything in terms of sin and cos first (using $\cot x = \frac{\cos x}{\sin x}$ rather than $\cot x = \frac{1}{\tan x}$)

Tip 2: Whenever you have algebraic fractions being added/subtracted, whether $\frac{a}{b} + \frac{c}{d}$ or $\frac{a}{b} + c$, combine them into one (as we can typically then use $\sin^2 x + \cos^2 x = 1$)

a. Simplify $\sin\theta\cot\theta\sec\theta$.

a

b. Simplify $\sin\theta\cos\theta(\sec\theta + \operatorname{cosec}\theta)$.

b

c. Prove $\frac{\cot\theta\operatorname{cosec}\theta}{\sec^2\theta + \operatorname{cosec}^2\theta} \equiv \cos^3\theta$.

c

## Test Your Understanding



1. Prove $\sec x - \cos x \equiv \sin x \tan x$.

1

2. Prove $(1 + \cos x)(\operatorname{cosec} x - \cot x) \equiv \sin x$.

2

## Solving Questions



Solve for $0 \leq \theta \leq 360°$:

a. $\sec \theta = -2.5$

b. $\cot 2\theta = 0.6$.

a

b

Solve $\cot \theta = 0$, $0 \leq \theta \leq 2\pi$.

c

c sketch

## Test Your Understanding



Solve in the interval $0 \leq \theta < 360°$:

$\operatorname{cosec} 3\theta = 2$

## Exercises 6C, Page 152



1. Rewrite as powers of $\sec\theta$, $\operatorname{cosec}\theta$ or $\cot\theta$.

a. $\frac{1}{\sin^3\theta}$

a

b. $\frac{4}{\tan^6\theta}$

b

c. $\frac{1}{2\cos^2\theta}$

c

d. $\frac{1-\sin^2\theta}{\sin^2\theta}$

d

e. $\frac{\sec\theta}{\cos^4\theta}$

e

f. $\sqrt{\operatorname{cosec}^3\theta\cot\theta\sec\theta}$

f

g. $\frac{2}{\sqrt{\tan\theta}}$

g

h. $\frac{\operatorname{cosec}^2\theta\tan^2\theta}{\cos\theta}$

h

2a. Find $\cot x$ if $5 \sin x = 4 \cos x$.

a

2b. Find $\cot x$ if $\tan x = -2$.

b

2c. Find $\cot x$ if $\frac{3 \sin x}{\cos x} = \frac{\cos x}{\sin x}$.

c



3. Simplify using the definitions of sec, cosec, cot and tan.

a. $\sin \theta \cot \theta$

a

b. $\tan \theta \cot \theta$

b

c. $\tan 2\theta \operatorname{cosec} 2\theta$

c

d. $\cos \theta \sin \theta (\cot \theta + \tan \theta)$

d

e. $\sin^3 x \operatorname{cosec} x + \cos^3 x \sec x$

e

f. $\sec A - \sec A \sin^2 A$

f

g. $\sec^2 x \cos^5 x + \cot x \operatorname{cosec} x \sin^4 x$

g

7. Solve for $0 \leq \theta \leq 2\pi$, giving answers in terms of $\pi$.

a. $\sec\theta = -1$

b. $\cot\theta = -\sqrt{3}$.

c. $\operatorname{cosec}(\theta/2) = 2\sqrt{3}/3$

d. $\sec\theta = \sqrt{2}\tan\theta,\ \theta \neq \pi/2, 3\pi/2$.

a

b

b diagram

c

d

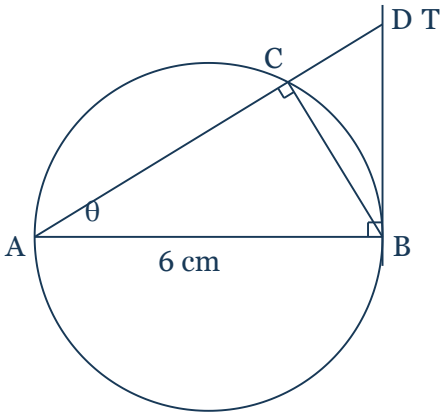
c diagram



8. $AB = 6$ cm is a diameter; $BT$ is tangent at $B$. Extend chord $AC$ to meet the tangent at $D$; $\angle DAB = \theta$.

a. Show $CD = 6(\sec\theta - \cos\theta)$ cm.

b. If $CD = 16$ cm, find $AC$.




a

b

9a. Prove $\frac{\operatorname{cosec} x - \cot x}{1 - \cos x} = \operatorname{cosec} x$.

a

9b. Hence solve $\frac{\operatorname{cosec} x - \cot x}{1 - \cos x} = 2$, $-\pi \leq x \leq \pi$.

b



10a. Prove $\frac{\sin x \tan x}{1 - \cos x} - 1 = \sec x$.

a

10b. Explain why $\frac{\sin x \tan x}{1 - \cos x} - 1 = -\frac{1}{2}$ has no solutions.

b

11. Solve $\frac{1+\cot x}{1+\tan x} = 5, 0 \leq x < 360°$.

11

## New Identities



From Year 1: $\sin^2 x + \cos^2 x = 1$.

Divide by $\cos^2 x$:

$1 + \tan^2 x = \sec^2 x$.

Divide by $\sin^2 x$:

$1 + \cot^2 x = \operatorname{cosec}^2 x$.

**Tip**

"One is tanned": 1 goes with $\tan^2 x$. Add "co" to both trig functions for the second identity.

Prove $1 + \tan^2 x \equiv \sec^2 x$.

Proof

**Tip**

This has been asked in an exam. Explicitly show each term divided by $\cos^2 x$.

## Examples



Prove $\operatorname{cosec}^4\theta - \cot^4\theta = \frac{1+\cos^2\theta}{1-\cos^2\theta}$.

Solve $4\operatorname{cosec}^2\theta - 9 = \cot\theta$, $0 \leq \theta \leq 360°$.

This is just like in C2; if you had say a mixture of $\sin\theta, \sin^2\theta, \cos^2\theta$: you'd change the $\cos^2\theta$ to $1 - \sin^2\theta$ in order to get a quadratic in terms of sin.

## Test Your Understanding



Edexcel C3 June 2013 (R), 6(ii). Solve $3\sec^2\theta + 3\sec\theta = 2\tan^2\theta$, $0 \leq \theta < 2\pi$. Show all working; give exact answers.

Solve $2\operatorname{cosec}^2 x + \cot x = 5$, $0 \leq x < 2\pi$, to 3 s.f.

2

7. Given $3\tan^2\theta + 4\sec^2\theta = 5$ and $\theta$ obtuse, find the exact value of $\sin\theta$.

7



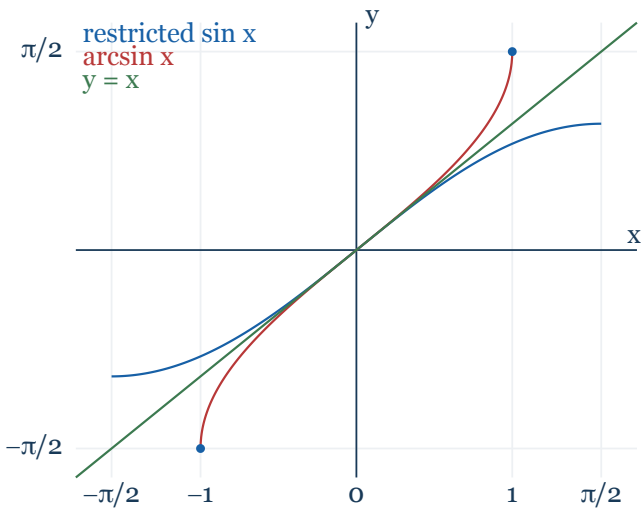
12. Given $2\sec^2\theta - \tan^2\theta = p$, show that $\operatorname{cosec}^2\theta = \frac{p-1}{p-2}$, $p \neq 2$.

12

You need to know how to sketch
$y = \arcsin x, y = \arccos x, y = \arctan x$.
(Yes, you could be asked in an exam!)

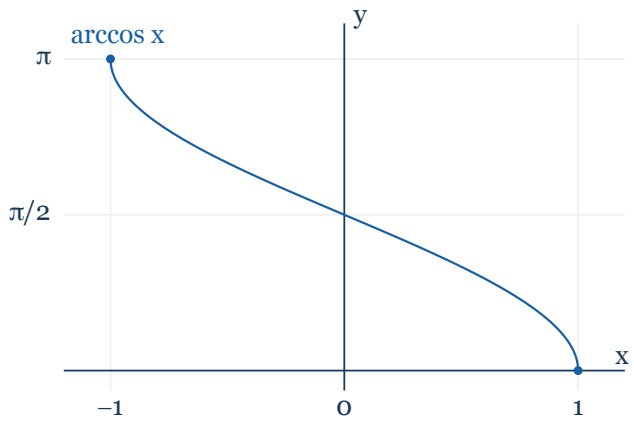
Before taking the inverse of sine, restrict its domain to $-\pi/2 \leq x \leq \pi/2$. Why?
Only one-to-one functions have inverse functions. The restriction makes sine one-to-one.
Reflect its graph in $y = x$.

For arcsin: domain $[-1, 1]$, range $[-\pi/2, \pi/2]$.

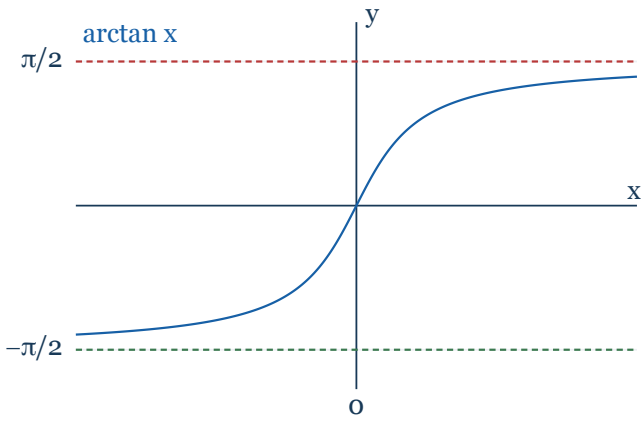





$y = \arccos x$
Domain $[-1, 1]$; range $[0, \pi]$.




$y = \arctan x$
Domain $\mathbb{R}$; range $(-\pi/2, \pi/2)$.
Horizontal asymptotes: $y = \pm\pi/2$.



## Evaluating inverse trig functions — Luisa · 39

[Textbook] Work out, in radians, the values of:

$\arcsin\left(-\frac{\sqrt{2}}{2}\right)$

$\arccos\left(-1\right)$

$\arctan\left(\sqrt{3}\right)$

a

b

You can simply use the $\sin^{-1}x$, $\cos^{-1}x$ and $\tan^{-1}x$ buttons on your calculator.

If you don't have a calculator, just use the sin, cos, tan graphs backwards.

c

## One Final Problem — Edexcel C3 January 2007 — Luisa · 40

Edexcel C3 January 2007, 8(ii).

Given $y = \arccos x$, $-1 \leq x \leq 1$, $0 \leq y \leq \pi$:

a. Express $\arcsin x$ in terms of $y$.

b. Hence evaluate $\arccos x + \arcsin x$, in terms of $\pi$.

Fewer than 10% of candidates got this part right.

a

b

3. Without a calculator, evaluate:

a. $\sin(\arcsin \frac{1}{2})$

b. $\sin(\arcsin(-\frac{1}{2}))$.

c. $\tan(\arctan(-1))$

d. $\cos(\arccos 0)$.

a

b

c

d



4a. Evaluate $\sin(\arccos(1/2))$ exactly, without a calculator.

4b. Evaluate $\cos(\arcsin(-1/2))$.

4c. Evaluate $\tan(\arccos(-\sqrt{2}/2))$.

a

b

c

b diagram

c diagram

5. Given $\arcsin k = \alpha$, $0 < k < 1$, with $\alpha$ in radians, write the first two positive solutions of $\sin x = k$ in terms of $\alpha$.

5



6. Given $\arcsin x = k$, $0 < k < \pi/2$:

a. State the possible values of $x$.

b. Express i. $\cos k$, ii. $\tan k$, in terms of $x$.

c. How do the answers to b change if $-\pi/2 < k < 0$?

a

b(i)

b(ii)

c

7. Sketch:

a. $y = \pi/2 + 2\arcsin x$ b. $y = \pi - \arctan x$ c. $y = \arccos(2x + 1)$.

a

b

c

b sketch

a sketch

c sketch



8. $f(x) = \arcsin x$, $-1 \leq x \leq 1$, and $g(x) = f(2x)$.

a. Sketch $y = f(x)$ and state the range of $f$.

b. Sketch $y = g(x)$.

a sketch

b

a range

b sketch